

Telecom Customer Churn Dashboard: Functional & Technical Requirements

Executive Dashboard Specifications & Layout Architecture by AI Business Analyst Advisory

TARGET PLATFORM: POWER BI / TABLEAU ENTERPRISE | AUDIENCE: EXECUTIVE LEADERSHIP & RETENTION OPERATIONS

1. EXECUTIVE SUMMARY & BUSINESS PURPOSE

This document specifies the end-to-end design requirements for an enterprise-grade **Telecom Customer Churn & Retention Dashboard**. Designed from an AI Business Analyst perspective, this dashboard bridges macro-level executive visibility with micro-level customer success operations, enabling proactive churn intervention before contract cancellation.

2. CORE BUSINESS QUESTIONS ADDRESSED

- **Macro Trend:** What is our overall monthly churn rate, and how is it tracking against target quarterly retention goals?
- **Segment Risk:** Which customer segments (contract type, tenure tier, payment method, fiber optic plans) exhibit the highest propensity to churn?
- **Financial Exposure:** What is the total Monthly Recurring Revenue (MRR) and Customer Lifetime Value (CLV) at immediate risk of defection?
- **Root Cause Drivers:** What are the primary predictive risk factors (e.g., support ticket escalation, billing spikes) driving individual account scores?

3. KEY PERFORMANCE INDICATORS (KPIs) & METRIC CARDS

KPI Name	Calculation Formula / Definition	Target / Benchmark	Visual Treatment
Total Active Subscribers	Count of active customer IDs at month-end	Growth vs. Prior Month	Big Number Card with MoM % Delta (Green/Red)
Monthly Churn Rate (%)	(Churned Customers in Month / Active Start Customers) × 100	< 2.0% Monthly	Gauge Chart / KPI Card with Warning Threshold
At-Risk MRR (\$)	Sum of Monthly Charges for customers with Churn Probability > 0.70	Minimize (\$)	Callout Card in Amber/Red Accent
Model Precision / Recall	Performance metrics of underlying AI Churn Prediction model	PR-AUC > 0.78	Small badge indicator on header

4. VISUAL DASHBOARD LAYOUT & WIREFRAME MOCKUP

The dashboard follows a 3-tier F-pattern layout optimized for executive scanning and operational filtering:

[TOP ROW: EXECUTIVE KPI CARDS & GLOBAL FILTERS]

[Active Subs: 7,043 (+1.2%)] | [Churn Rate: 26.5% (-0.8%)] | [At-Risk MRR: \$138.4K] | [Model AUC: 0.84] [Filters: Region | Tenure | Contract]

[MIDDLE ROW: TREND ANALYSIS & RISK DRIVERS]

[1. Churn Rate Trend Over Time (Line Chart)] [2. Churn by Contract & Payment Method (Stacked Bar)]

[BOTTOM ROW: GEOSPATIAL / SEGMENT MATRIX & DRILL-THROUGH TABLE]

[3. Feature Importance / SHAP Drivers (Bar)] [4. High-Risk Customer Action Table (Drillable)]

5. FILTERS & SLICERS

- **Global Slicers:** Date Range (Month/Quarter/Year), Region / Territory, and Customer Tier.
- **Interactive Toggles:** Contract Type (Month-to-Month vs. 1-Yr vs. 2-Yr), Payment Method, and Internet Service Type.
- **Risk Score Slicer:** Dynamic slider allowing users to filter accounts by AI Churn Probability (> 0.50, > 0.80).

6. TOOLTIP & DRILL-THROUGH SPECIFICATIONS

- **Hover Tooltips:** Hovering over any segment in the bar chart reveals a mini-card showing Average Tenure, Total Charges, and Top 2 SHAP Risk Factors.
- **Drill-Through Target:** Right-clicking any customer ID or segment in the summary table drills through to the "Customer 360 Risk Profile Page" displaying individualized retention offers and support history.

7. COLOR THEME & PROFESSIONAL FORMATTING

Adheres to a professional "Slate & Deep Navy" financial analytics palette designed to minimize cognitive fatigue and highlight actionable risk alerts:

- **Base / Background:** Off-white / light slate (`#F8FAFC`) with clean white (`#FFFFFF`) card containers.
- **Headers & Structural Text:** Deep Slate Navy (`#0F172A`) for absolute clarity.
- **Accent / Interactive Elements:** Electric Cyan / Teal (`#06B6D4`) for primary highlights and active states.
- **Risk Alert Palette:** Desaturated Amber (`#F59E0B`) for moderate risk and Coral Red (`#EF4444`) strictly reserved for high-propensity churn metrics.

8. NAVIGATION & PAGE ARCHITECTURE

- **Page 1: Executive Overview** — High-level KPIs, monthly trends, and revenue exposure summary.
- **Page 2: Churn Driver Deep Dive** — Interactive SHAP feature importance, contract correlation matrices, and behavioral breakdowns.
- **Page 3: Customer Retention Operations (Action Hub)** — Actionable table of high-risk accounts with integrated CRM ticket triggers and one-click discount authorization.